Recitation-6

Ridge Regression:

L1 Regularization.

property features & prices

Bedrooms x_1	Age(years) x_2	price (k 100) y	$y-\hat{y}$
3	5	2	-0.37
3	3	3	-0.38

$y = mx + c$

$2 = c + 3m_1 + 5m_2$

$3 = c + 3m_1 + 3m_2$

$$\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 & 3 & 5 \\ 1 & 3 & 3 \end{pmatrix} \begin{pmatrix} c \\ m_1 \\ m_2 \end{pmatrix}$$

$\lambda = 1$

$$w = (X^T X + \lambda I)^{-1} (X^T y)$$

Regularization

$$\lambda = 1, \quad I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times 3}$$

$$X^T X = \underset{3\times 2}{\begin{pmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{pmatrix}} \underset{2\times 3}{\begin{pmatrix} 1 & 3 & 5 \\ 1 & 3 & 3 \end{pmatrix}}$$

$$\begin{pmatrix} 2 & 6 & 8 \\ 6 & 18 & 24 \\ 8 & 24 & 34 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 6 & 8 \\ 6 & 19 & 24 \\ 8 & 24 & 35 \end{pmatrix}$$

$$X^T \begin{pmatrix} 3 & 6 & 8 \\ 6 & 19 & 24 \\ 8 & 24 & 35 \end{pmatrix}^{-1} = \frac{1}{95} \begin{pmatrix} 89 & -10 & -8 \\ -18 & 41 & -24 \\ -8 & -24 & 21 \end{pmatrix}$$

$$X^T y = \begin{pmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 15 \\ 19 \end{pmatrix}$$

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$$\begin{pmatrix} 23/95 \\ 69/95 \\ -1/95 \end{pmatrix} = \begin{pmatrix} 0.242 \\ 0.727 \\ -0.011 \end{pmatrix} \begin{matrix} \to c \\ \to m_1 \\ \to m_2 \end{matrix}$$

$$\hat{y}_1 \text{pred} = 0.2 + 0.7(3) - 0.01(5)$$

$$= 2.38$$

$$\hat{y}_2 = 2.38.$$

$$MSE = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2$$

$$= 0.25$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{N} (y_i - \bar{y})^2} \Rightarrow -0.842.$$